

GENERAL SPECIFICATION G7.3

Testing and Commissioning (ACMV)

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G7.3 TESTING AND COMMISSIONING (ACMV)

G7.3.1 Testing Apparatus

- (a) The Contractor shall provide all necessary testing equipment to carry out the complete balancing, testing and commissioning of the installation. The minimum testing equipment he is expected to have are set out below:

Unit(s)	Equipment
1	- Combined mounted maximum and minimum thermometer
1	- Velometer with accessories and case
1	- Vane anemometer with case
1	- Recorder (with daily/weekly charts) for temperature and humidity
1	- Mounted °C thermometer
1	- Mounted °C sling psychrometer
1	- Sound level meter with octave-band frequency analyser
2	- Water Pressure Gauge
1	- Any other equipment required by the manufacturer's instructions

- (b) The Contractor shall not include the full cost of these testing equipment in the Total Tender Price, as these testing equipment will remain his property.
- (c) All testing apparatus as well as measuring instruments, gauges, recorders, etc., which are specified and installed under this Contract, shall be calibrated to the S.O.'s satisfaction. Calibration certificates shall be submitted to the S.O.. All testing apparatus shall be stamped with approved stickers showing the validity of calibration.
- (d) All equipment must comply with SS591.

G7.3.2 Fan Coil Units, Fans and Ductwork

- (a) Fan coil units and fans shall be checked over to ensure filters are clean and all in position, and that no air is being 'by-passed'. All duct connections shall be checked for air tightness before balancing is commenced. Air quantities shall be set to within designed figures plus 10% tolerance as given on the Drawings for air quantity less than 17000 CMH and a reduction of 1% for each increment of 17000 CMH. Air quantity above 170,000 CMH shall be set to within 1% of the figures given on the drawings and final testing being done by spot checking of temperature throughout each zone.
- (b) Each ventilation fan shall be checked for rotational speed, air volume, temperatures and pressure drop; these figures shall be recorded and submitted in duplicate to the S.O.. Outside air dampers shall be balanced to meet the design requirement to within 10% of the designed air quantity. For modulating outside air dampers, balancing shall be carried out with the dampers set in a fully open position to ensure that the peak design outside air quantities are achieved.

- (c) After balancing is completed, each damper quadrant and adjusting arm shall be printed in red with matching indicating marks and firmly locked in position.

G7.3.3 Air Outlets

- (a) The methods of balancing of air outlets shall be in accordance with a recognised standard for field measurement and instrumentation, and shall be to the approval of the S.O..
- (b) Vane anemometer may be used for air grilles used in conjunction with mechanical ventilation systems, or for side discharge grilles. The average anemometer reading shall be determined by marking the grille off in sections, taking the readings in front of each section and averaging the results. Moving the anemometer back and forth across the face of the grille to obtain the average reading shall not be permitted without the approval of the S.O..
- (c) For air diffusers used in conjunction with the air conditioning systems, the air quantities shall be measured with the use of proprietary portable measuring stations. Suitable collecting sections or cones shall be used to facilitate measurements to be readily taken. Air flow meters shall be calibrated before use.
- (d) All grilles/diffusers shall be set to within designed figures plus the same tolerance as specified for the air-handling units and ductwork.

G7.3.4 Control Equipment

- (a) Contractor shall calibrate all thermostats and humidistats, set the modulating range and set points on all automatic valves and dampers as required to ensure that operating directions are correct. The time and control sequence shall also be tested and verified.

G7.3.5 Refrigeration Equipment

- (a) Each refrigeration compressor shall be tested in accordance with the manufacturer's instructions. The pressure control settings shall be adjusted and set. The capacity control mechanisms shall be verified and the lead/trailing and stage loading operation and sequencing shall be checked and confirmed.
- (b) Each water chilling unit shall be checked, tested and recorded under design conditions; performance tests shall comply with latest appropriate ARI Standards or approved equivalent standards.

G7.3.6 Pipework

- (a) All piping and fittings used shall have been tested in accordance with the relevant BS code or specification. In addition, the pipe pressure testing shall be done during or after the erection of the piping system and before pipes are lagged or covered over.

- (b) For the refrigerant piping, the method of test shall be to pump the system to the required test pressure which shall be maintained with all valves shut -for a period of 24 hours or for such time as is necessary to carefully inspect all joints and valves, whichever is the longer. When a gas is used for testing, all joints shall be inspected after application of soapy work. In addition, piping shall be tested at working pressure during operation of the system through several cycles of temperature pressure and varying conditions of operation until the S.O. is satisfied that the piping will remain tight, sound and true to line and not damage itself or other structure.

G7.3.7 Refrigerant Piping

- (a) Piping shall be tested to pressure as recommended by the manufacturers using anhydrous carbon dioxide or nitrogen. After charging with refrigerant, the system shall be thoroughly checked for leaks, using the Halide torch principle.
- (b) All pressure tests of pipework shall be to the approval of the S.O. before any insulation of pipework may commence.

END OF SECTION